It's More Than Time and Money: How the Parent-Child Relationship Shapes Skill Formation

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Motivation

- Economists model child development as shaped by parents' **time and money** investments
  - But effects on socio-emotional (SE) skills are mixed
- Child psychologists emphasize the **parent-child bond**—the emotional connection between parent and child
  - Stronger bonds predict better child outcomes
  - Largely absent from economics, mainly due to a lack of data
- **This paper:** bridges the two literatures with a skill-development model which explicitly accounts for parent–child bond
  - Identified using high-quality Canadian administrative data

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What Is the Parent–Child Bond?

Psychologists define the parent–child bond as the emotional connection between child and caregiver:

- **Built through** repeated interactions (nurturance, rejection, monitoring)
- **Seen in** the child's sense of closeness, security, and trust

(Schaefer, 1965; Roberts et al., 1984; Lempers et al., 1989; Stattin & Kerr, 2000)

Implication: the bond is malleable—parents can invest in it, particularly by spending time with their children

This Paper

1. **This paper:** A dynamic structural model of two-parent household decisions
   - Parents choose work, leisure, childcare time, consumption, and child spending
   - Time and spending combine into investment, which builds the bond and skills
   - The bond enters the parent's utility directly, not just as a route to child skills

2. **Novel data:** Exploit Canadian data with repeated measures on:
   - Parent-child bond (18) and socio-emotional skills (29),
   - Parental labour supply, and income,
   - Linked to administrative adult earnings on the children aged 10 to 15

3. **Approach:** I build a Simulated Method of Moments estimator that enables me to estimate the model parameters across four Canadian datasets
   - Time Diaries: trade-offs between work, leisure, and child time
   - Household Spending Surveys: consumption and child spending

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Contributions and Findings

1. **Technology of child skill development**

   (Cunha & Heckman, 2008; Cunha et al., 2010; Attanasio et al., 2020; Wiswall & Agostinelli, 2024)

   - The bond drives future skill; investment matters only by building the bond
   - Bond and skill are complements: a stronger bond helps higher-skill children more

2. **Intrahousehold allocation of time and money**

   (Del Boca et al., 2014; Blundell et al., 2018; Agostinelli et al., 2024)

   - I endogenize the bond in a dynamic household model of time and money allocation
   - Income transfers barely move children's skill
   - Parents mostly spend freed resources on their own consumption and leisure

3. **Policy impact of parenting programs**

   (Kim et al., 2018; Sanders et al., 2014)

   - Raising the parent–child bond increases skill far more than income transfers do
   - Raising the bond is about as effective as raising child skill directly

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Data Sources and Key Variables: Four Canadian Datasets (Part 1)

NLSCY (1994–2009) Longitudinal; children aged 10–15, 3 age bins (10–11, 12–13, 14–15); two-parent households only

1. **Latent parent-child bond (P_t):** Nurturance (7) + Rejection (7) + Parental Knowledge (4), child-reported **(new)** ▸ Items
2. **Latent socioemotional skills (θ_t):** inattention, emotion, aggression, prosocial (29 items), child-reported ▸ Items
3. **Parental Labour:** annual wage & salary earnings; usual weekly hours; household income
4. **Latent investment (I_t):** parental school participation, active parenting time, outside-school activities (13 items) ▸ Items
5. **T1FF (1993–2019):** NLSCY children linked to own adult earnings records

Measuring the Bond

The bond is not observed directly. I measure it from 18 child-reported items on three dimensions of parental interaction (Lempers et al., 1989; Stattin & Kerr, 2000; Del Bono, Kinsler & Pavan, 2024; Barszczewski & Thomas, 2025)

Nurturance (warmth, support)

- listen to my ideas and opinions
- seem proud of the things I do

Rejection (inconsistent, punitive discipline)

- enforce rules depending on their mood
- get angry and yell at me

Parental Knowledge (disclosure, trust/openness)

- know where I am and what I'm doing
- tell me what time to be home

Higher nurturance and knowledge, lower rejection ⇒ higher bond. ▸ Full Bond Measures

Data Sources and Key Variables: Four Canadian Datasets (Part 2)

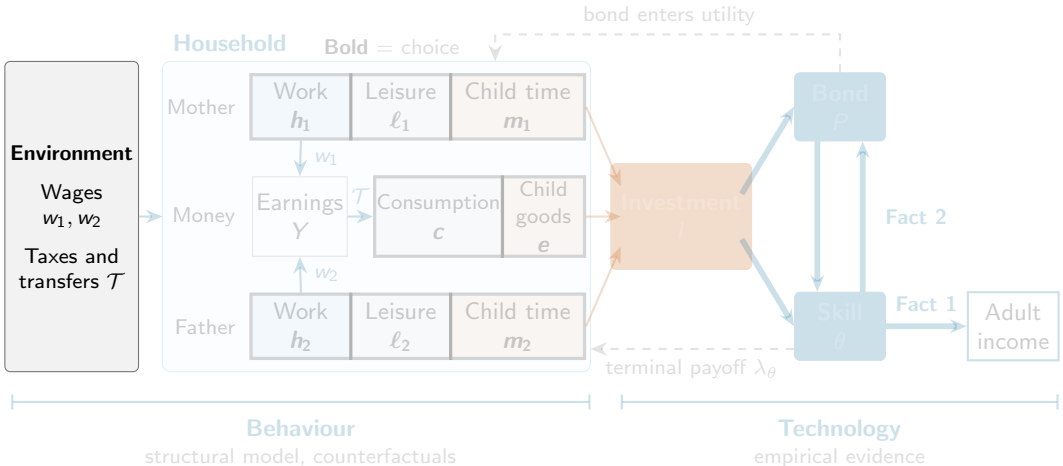
1. **GSS Time Use (2005):** time diaries → market work, active parenting time (homework, play, supervision), residual (leisure)
2. **SHS (1997–2009):** child-related spending → clothing & footwear, toys, education (tuition, supplies, extracurriculars)

Main sample. Two-parent households, at least one biological parent present; both parents with non-missing hours, employment status, income.

- 3,190 children, observed in two or three age bins (9,580 observations).
- Bond (P), Skill (θ), and Investment (I) are latent factors from child-reported survey items.

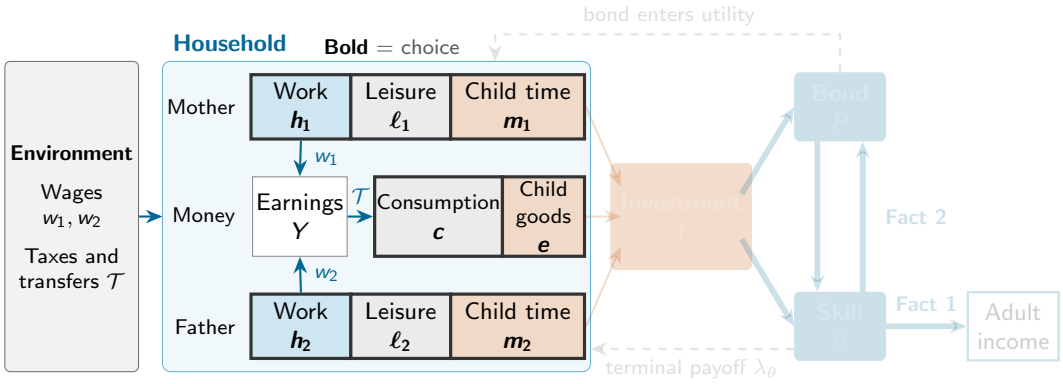
▸ Latent Factor Measurement

Overview: How Time and Money Reach the Child



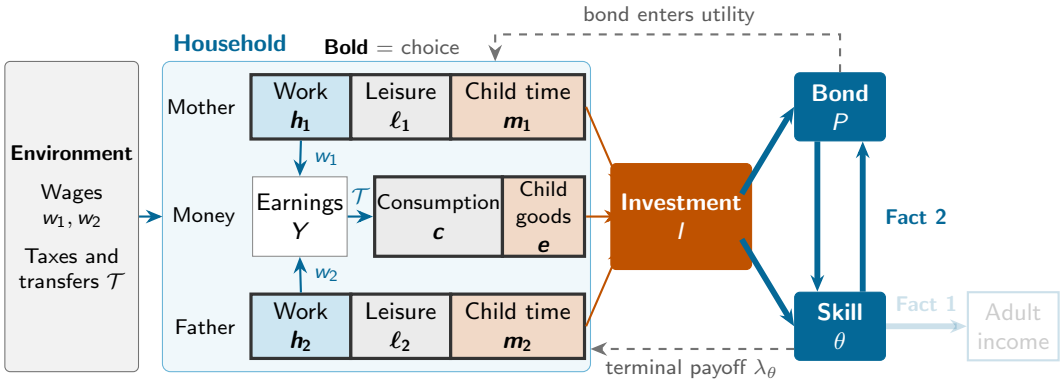
Wages and the tax-and-transfer schedule are the environment.

Overview: How Time and Money Reach the Child



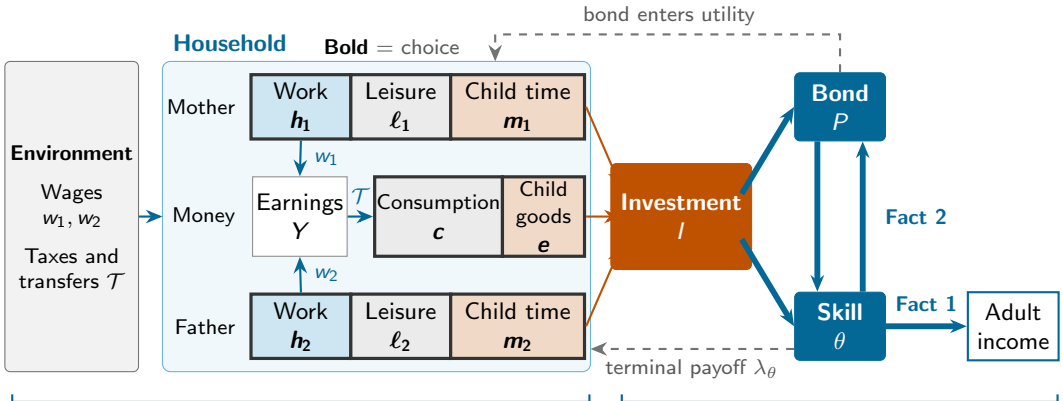
Parents choose time and money.

Overview: How Time and Money Reach the Child



Investment builds the bond, and the bond builds skill.

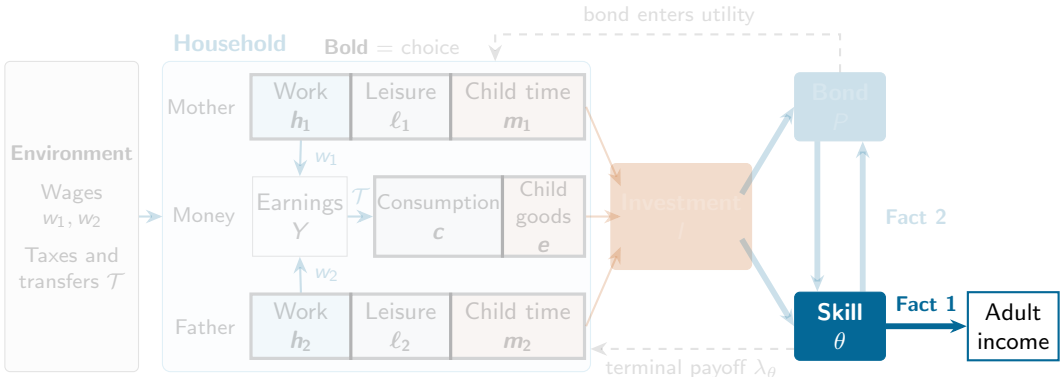
Overview: How Time and Money Reach the Child



Skill at ages 14–15 predicts adult income.

Fact 1

Adolescent socio-emotional skill predicts adult income



Adolescent SE Skill Predicts Adult Income

Adult log income (25-36 years of age) on adolescent skill, math, and the bond (Bartlett-scored, Croon-corrected, per SD)

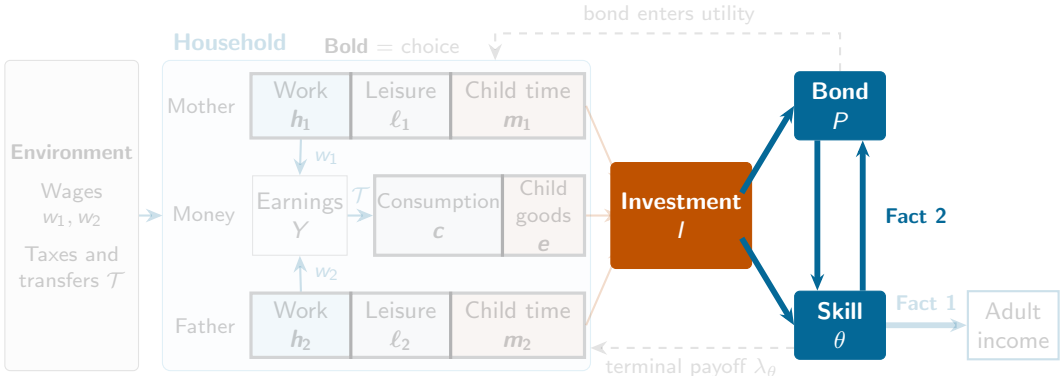
$$\ln \text{Income}_{i,t} = \alpha + \beta_1 \ln \theta_{i,14\text{–}15} + \beta_2 \text{Math}_{i,14\text{–}15} + \beta_3 \ln P_{i,14\text{–}15} + Z_i\gamma + \lambda_t + \mu_c + \varepsilon_{i,t}$$

	[1]	[2]	[3]	[4]	[5]
Child SE Skill (θ)	0.114***	0.106***	0.083**	—	0.072*
	(0.042)	(0.038)	(0.035)		(0.037)
Parental Bond (P)	—	—	—	0.025**	0.009
				(0.010)	(0.011)
Math Score (standardised)	—	—	0.097***	—	0.096***
			(0.014)		(0.014)
Log Household Income (14–15)	—	0.082***	0.075***	0.081***	0.074***
Year, cohort FE			✓		
Male, sibs., mom age, parent educ.	—	✓	✓	✓	✓
N (individuals / person-years, rounded)			3,980 / 14,800		

▸ Age 30+

Fact 2

Investment builds the future bond → the bond builds future skill.



Specification: Production Functions

Cunha–Heckman-style Translog production functions for skill and bond:

$$\ln \theta_{i,t+1} = \beta_1^\theta \ln \theta_{i,t} + \beta_2^\theta \ln P_{i,t} + \beta_3^\theta \ln I_{i,t} + \beta_4^\theta \ln \theta_{i,t} \ln P_{i,t} + \beta_5^\theta \ln I_{i,t} \ln \theta_{i,t} + X_{i,t}\beta_X^\theta + \varepsilon_{i,t}^\theta$$

$$\ln P_{i,t+1} = \beta_1^P \ln P_{i,t} + \beta_2^P \ln \theta_{i,t} + \beta_3^P \ln I_{i,t} + \beta_4^P \ln \theta_{i,t} \ln P_{i,t} + \beta_5^P \ln I_{i,t} \ln P_{i,t} + X_{i,t}\beta_X^P + \varepsilon_{i,t}^P$$

θ: skill; P: bond; I: investment; X: family and child controls. β_4: skill $\times$ bond; β_5: investment $\times$ own lag. Cobb–Douglas sets $\beta_4 = \beta_5 = 0$.

Latent factors are corrected for measurement error and standardised (SD units). ▸ details

Reduced-Form Production Functions

	Cobb-Douglas		Translog	
	SE Skill **[1]**	Parental Bond **[2]**	SE Skill **[3]**	Parental Bond **[4]**
Investment	0.021	0.110***	0.022	0.108***
	(0.019)	(0.025)	(0.019)	(0.024)
SE Skill (lag)	0.605***	−0.032	0.662***	0.026
	(0.049)	(0.046)	(0.045)	(0.043)
Parental Bond (lag)	0.100*	0.675***	0.163***	0.743***
	(0.052)	(0.069)	(0.051)	(0.074)
Investment × own lag	—	—	0.014	−0.005
			(0.017)	(0.018)
Skill × Bond (lag)	—	—	0.089***	0.097***
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Controls			✓	
Observations (child-transitions)			3,480	

▸ Mediation

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Why We Need a Structural Model

I showed that:

- Investment's production on skill $\longrightarrow$ **small** Bolt et al. (2021); Attansio et al. (2020); Mitchell et al. (2023)
- Investment $\longrightarrow$ bond $\longrightarrow$ skill.
- Bond is a mechnism in which investment builds skill. Akee et al. (2018)

Reduced form cannot tell us:

- How households would reoptimize/change their behavior under constraints they have not faced (a transfer, a change to the tax schedule)
- What this implies for the bond and skill, and thus adult income

For that we need a structural model of household decision-making.

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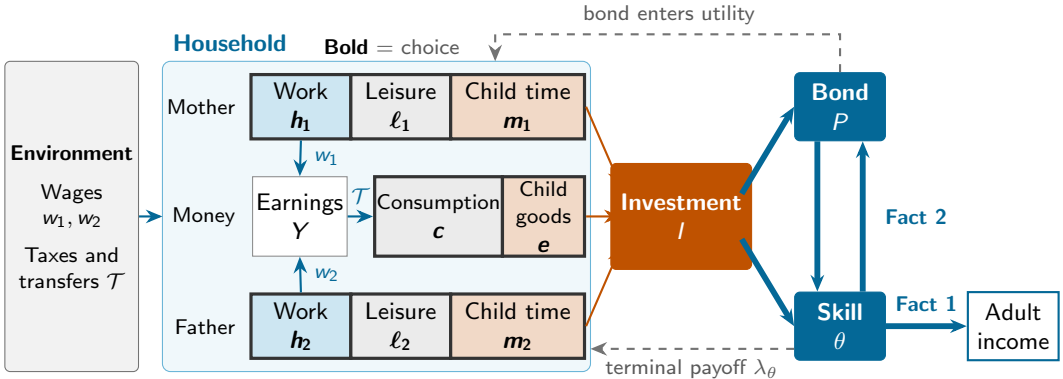
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A Dynamic Model of Household Decisions and the Parental Bond



Household's Problem

A unitary two-parent household maximizes utility over childhood ($t = 0, \ldots, T$, ages 10–15), with a terminal payoff at $T + 1$.

$$V_t(S_t) = \max_{x_t \in \Gamma(S_t)} \left\{ u_t(c_t, \ell_{1t}, \ell_{2t}, P_t) + \beta \, \mathbb{E}[V_{t+1}(S_{t+1}) \mid S_t, x_t] \right\}$$

- Controls $x_t = (c_t, e_t, h_{1t}, h_{2t}, m_{1t}, m_{2t})$
- States $S_t = (\theta_t, P_t, w_{1t}, w_{2t})$
- Subscript 1 = mother, 2 = father

c_t	consumption
e_t	child expenditure
ℓ_{jt}	leisure, parent j
h_{jt}	hours worked
m_{jt}	active time with child
θ_t	child SE skill
P_t	parent–child bond
w_{jt}	wage offer, parent j

Terminal value (Del Boca et al. 2014):

$$V(\theta_{T+1}) = \lambda_\theta \ln \theta_{T+1}$$

▸ Terminal Value ▸ Utility

Constraints

$$
\begin{aligned}
c_t + e_t &= \mathcal{T}(Y_t) && \text{(Budget)} \\
\mathcal{T}(Y_t) &= \mathcal{X}_1(\mathcal{X}_2 + Y_t)^{1-\mathcal{X}_3} && \text{(Tax and transfer, CTaCS Milligan, 2007)} \\
Y_t &= w_{1t}h_{1t} + w_{2t}h_{2t} && \text{(Earnings)} \\
\ell_{jt} + h_{jt} + m_{jt} &= 1, \quad j \in \{1,2\} && \text{(Time)} \\
\theta_{t+1} &= f(\theta_t, P_t, I_t) && \text{(Skill Tech., translog)} \\
P_{t+1} &= g(P_t, \theta_t, I_t) && \text{(Bond Tech., translog)} \\
I_t &= \mathcal{I}(e_t, m_{1t}, m_{2t}) && \text{(Investment bundle)}
\end{aligned}
$$

No saving $\Rightarrow$ Del Boca et al., 2014, 2025; Agostinelli et al., 2024

▸ Why Invest in the Bond? ▸ Technologies

The Investment Aggregator

$$\ln I_t = \alpha_e \ln e_t + \alpha_{m_1} \ln m_{1t} + \alpha_{m_2} \ln m_{2t} + \alpha_{12} \ln m_{1t} \ln m_{2t}$$

- α_e, α_{m_1}, α_{m_2}: elasticities w.r.t. spending, parents' time
- α_{12}: maternal/paternal time **complements** (> 0) or **substitutes** (< 0)

Wage Process: Del Boca et al. 2025

For each parent $j \in \{1, 2\}$, log wage offer in age bin t:

$$\ln w_{jt} = b_{0j} + b_{1j}\,\mathrm{edu}_j + b_{2j}\,\mathrm{age}_{jt} + \varepsilon_{jt}$$

with i.i.d. jointly normal shocks correlated across spouses:

$$\begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2_{\mathrm{mom}} & \rho_{\mathrm{wage}}\sigma_{\mathrm{mom}}\sigma_{\mathrm{dad}} \\ \rho_{\mathrm{wage}}\sigma_{\mathrm{mom}}\sigma_{\mathrm{dad}} & \sigma^2_{\mathrm{dad}} \end{pmatrix} \right)$$

- ρ_{wage}: assortative mating, shared local labour market
- Serially uncorrelated $\Rightarrow$ wage risk is a static reallocation within each age bin
- $(b_{0j}, b_{1j}, b_{2j}, \sigma_j, \rho_{\mathrm{wage}})$ estimated jointly inside the SMM

Two-Step Strategy

Step 1: outside the model

- Discount factor $\beta = 0.96^2$ (two-year periods).
- Tax-and-transfer parameters $(\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3)$: NLS on the Canadian Tax and Credit Simulator (CTaCS).
- Initial (θ_0, P_0): drawn from initial distribution estimated from the NLSCY.

Step 2: inside the model (SMM), 30 parameters, 93 moments

- Preferences $\{\alpha_c, \alpha_P, \alpha_{\ell_1}, \alpha_{\ell_2}\}$, skill terminal λ_θ, and investment aggregator $\{\alpha_e, \alpha_{m_1}, \alpha_{m_2}, \alpha_{12}\}$
- Skill technology $\{a, \delta_\theta, \gamma_1, \gamma_2, \gamma_3, \gamma_4, \sigma_\varepsilon^\theta\}$ and bond technology $\{\tilde{a}, \delta_P, \tilde{\gamma}_1, \tilde{\gamma}_2, \tilde{\gamma}_3, \tilde{\gamma}_4, \sigma_\varepsilon^P\}$
- Wage process $\{b_{0j}, b_{1j}, b_{2j}, \sigma_j\}_{j=1,2}$, ρ_{wage}

▸ SMM Algorithm ▸ Identification

Results

Parameter Estimates: Preferences & Investment

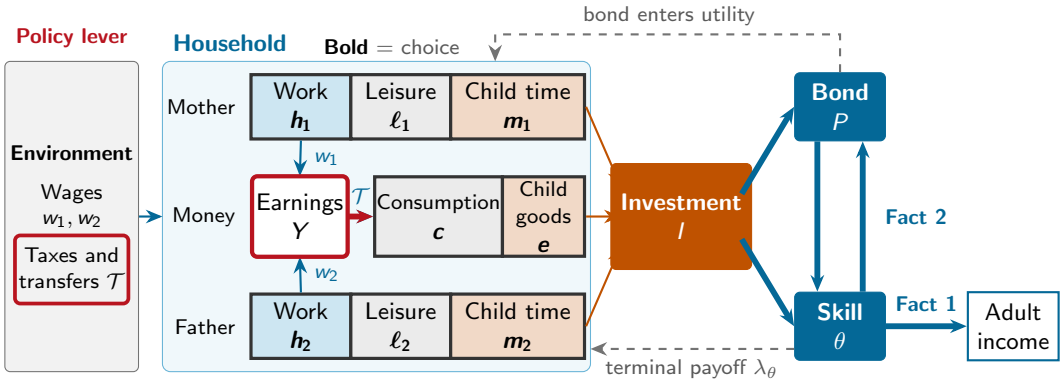
	Description	Estimate	SE
Preferences and terminal			
α_P	Bond utility weight	0.122	0.0012
α_{ℓ_1}	Mom leisure weight	0.256	0.0006
α_{ℓ_2}	Dad leisure weight	0.282	0.0005
α_c	Consumption weight (implied)	0.340	—
λ_θ	Skill terminal weight	2.781	0.0168
Investment technology			
α_e	Expenditure elasticity (implied)	0.227	—
α_{m_1}	Mom time elasticity	0.461	0.0053
α_{m_2}	Dad time elasticity	0.311	0.0028
α_{12}	Mom $\times$ dad time interaction	0.022	0.0006

- **Consumption $>$ either parent's leisure**, but total leisure $>$ consumption.
- **Fathers' leisure weighted more** than mothers'.
- **Mothers' time carries the largest weight** in investment, then fathers', then goods.
- **Mom $\times$ dad time interaction** is small and slightly positive.
- **Technology estimates** are in latent units. The counterfactual ATEs interpret them.
  (Del Bono et al., 2022; Freyberger, 2026)

▸ Technology Estimates ▸ Latent Units ▸ Why Invest? ▸ Model Fit

Policy Counterfactuals: Taxes and Transfers

Reforms change $\mathcal{T}$, the map from earnings Y to after-tax income



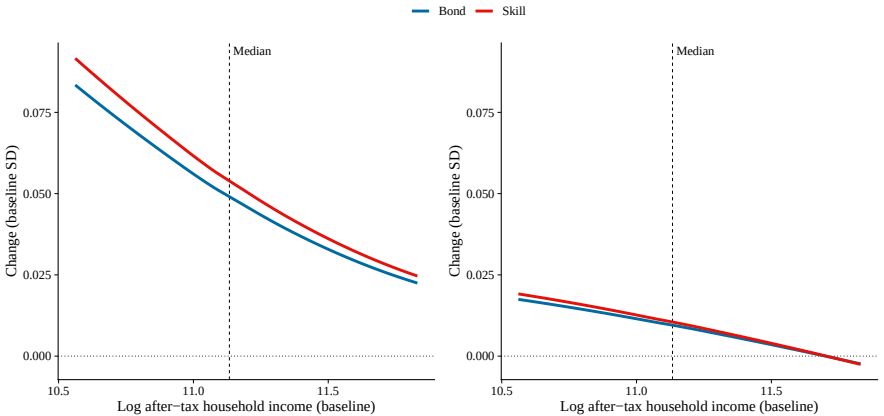
Policy Counterfactuals

After-tax income: $\mathcal{T}(Y_t) = \mathcal{X}_1(\mathcal{X}_2 + Y_t)^{1-\mathcal{X}_3}$.

1. Income Transfer Raise the benefit floor $\mathcal{X}_2$ by **$24,000** (Yang's $1,000/month proposal), with additional resources concentrated on low-income families.

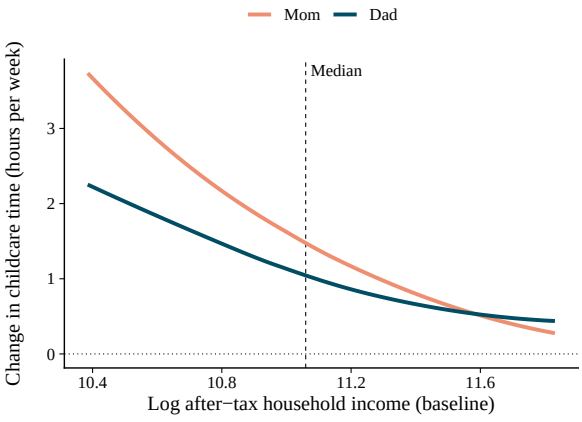
2. More progressivity, same burden. Double $\mathcal{X}_3$ and readjust $\mathcal{X}_1$ so the median household's average tax rate stays at 21.6%.

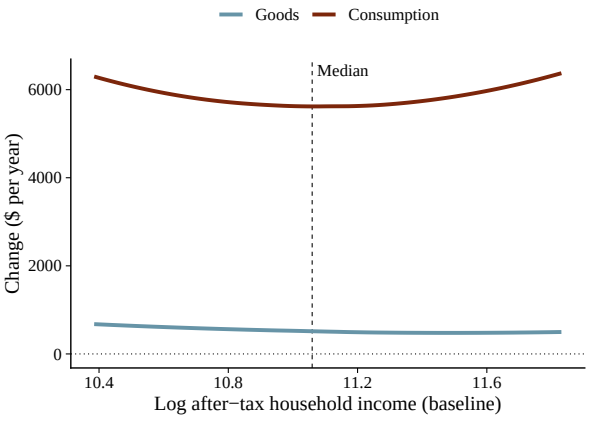
▸ Mechanical mapping



- UBI (left): gains everywhere, largest at the bottom. Skill moves $\sim 1.1\times$ as much as the bond; gap narrows going up; Progressivity (right): small gains everywhere, near zero at the top.

▸ Marginal rate fixed





- Gains driven by the time channel, not money. UBI buys time.

▸ Work & Leisure ▸ Progressivity

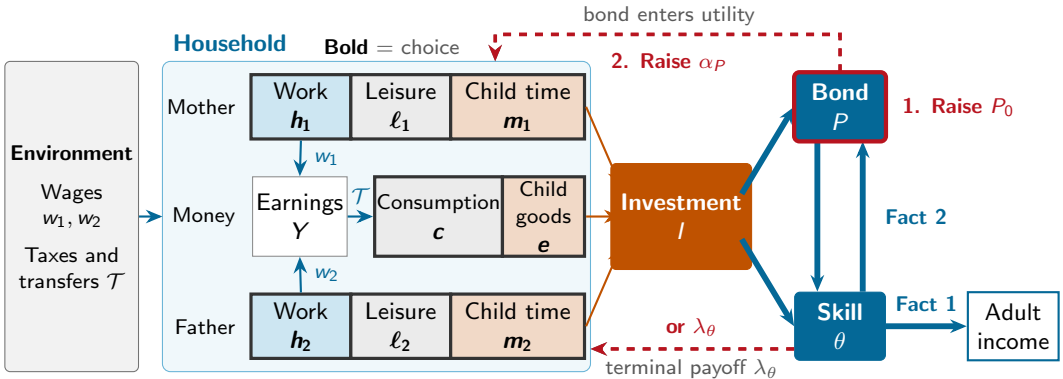
Policy Counterfactuals: Parenting Program

Example: Triple P

- 72 countries, 22 languages, 7M+ families reached, 100,000 practitioners trained, and 690 studies (349 clinical trials)
- Teaches consistent discipline, warmth, and realistic expectations.
- Randomized trial: parents attend four weekly two-hour group sessions. No money, little time. (Kim et al., 2018)
- Harsh discipline falls and positive engagement rises 0.2-0.4 SD 4 years after the intervention; at ages 12–16, children's externalizing behaviour is 0.31 SD lower.
- Benefits are not just for low SES families: the program works across the income distribution.

Policy Counterfactuals: Bond and Parental Values

A parenting program can raise the bond P_0 or change what parents value



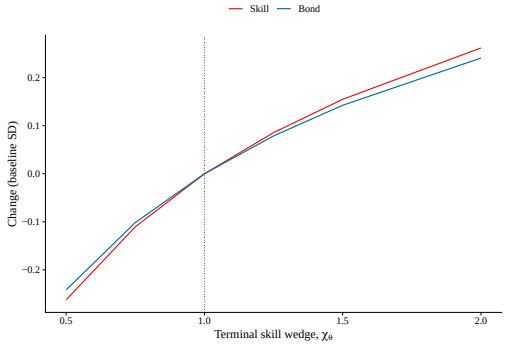
Policy Counterfactuals: Parenting Program Results

	Transfer	Stronger start (ages 10–11)		Parents' weight +50%		
Change at ages 14–15	UBI $24k	$\theta_0 \uparrow 1$ SD	$P_0 \uparrow 1$ SD	α_P	χ_θ	Both
Child skill (SD)	0.055	**0.409**	**0.425**	0.050	0.155	0.197
Parental bond (SD)	0.050	0.118	**0.418**	0.045	0.143	0.181
Adult income	0.96%	7.42%	7.73%	0.89%	2.75%	3.51%

- A stronger bond does as much for skill as stronger skill, **and** raises the bond far more.
- Parents who value the bond or skill more invest more, mostly out of leisure.
- Both weights together raise skill by less than the sum of the two.
- All but the bond weight raise skill more than the UBI (not cost-matched).

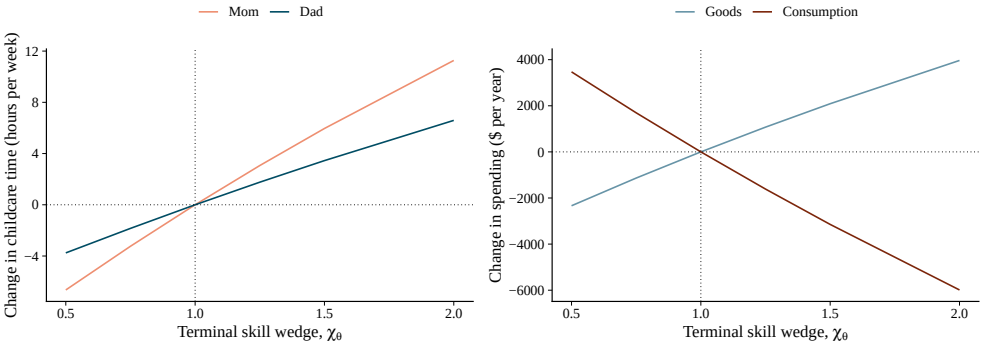
▸ Triple P Table

Mechanisms: Varying the Weight on the Child's Future



- Doubling χ_θ raises skill by $\sim$**0.26 SD** and the bond by $\sim$**0.24 SD**; halving it lowers them by the same amounts. Skill moves slightly more than the bond.
- **Concave:** -50% lowers skill by 0.262 SD, $+50\%$ raises it by only 0.155 SD.

Weight on the Child's Future: Childcare and Spending



- Doubling χ_θ: childcare +11 hrs/wk for moms, +6.6 for dads. Spending shifts from consumption to child goods.

Takeaways

The parent–child bond, not investment itself, is at the centre of SE skill formation.

- The parent-child bond is a mechnism in which household resources can reach the child SE Skill.
- The bond raises skill more than skill raises the bond.
- Income transfers do little for skill; programs that **strengthen the bond**, or change how parents value it, may do more.

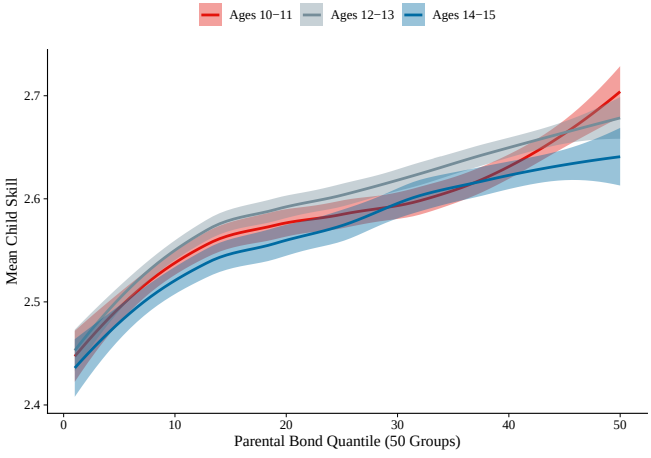
▸ Full Results Table ▸ By Income Quartile ▸ Cash Equivalent
▸ Relationship Motive ▸ Weight on Child's Future

Thank you

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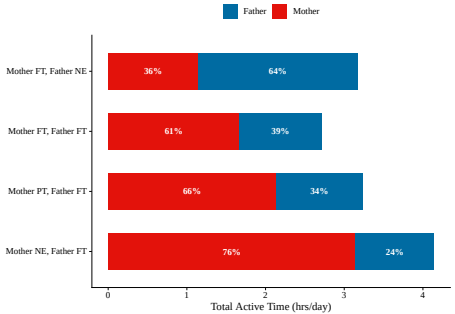
The Bond and Child Skill Move Together

Back

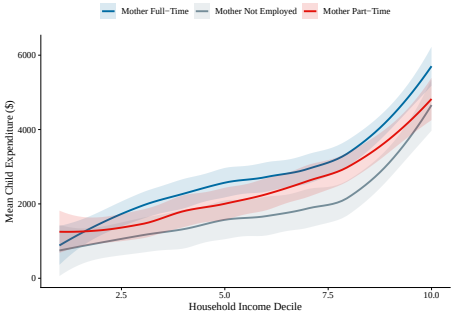


Time and Money Track Household Circumstances

Back



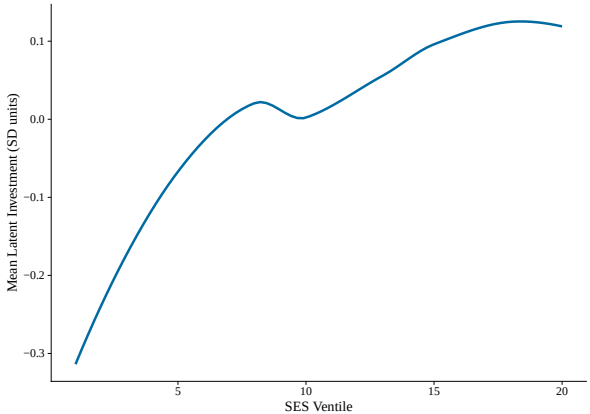
Active time and its division, by couple employment type (GSS).



Child expenditure by household income decile (SHS).

Investment Rises with Socioeconomic Status

Back



Robustness: Restricting to Age 30+

◂ Back **Adult log income (30+ years of age) on adolescent skill, math, and the bond (Bartlett-scored, Croon-corrected, per SD)**

	[1]	[2]	[3]	[4]	[5]
Child SE Skill (θ)	0.140**	0.130**	0.097*	—	0.089
	(0.064)	(0.059)	(0.054)		(0.057)
Parental Bond (P)	—	—	—	0.029	0.008
				(0.018)	(0.018)
Math Score (standardised)	—	—	0.144***	—	0.144***
			(0.024)		(0.024)
Log Household Income (14–15)	—	0.083*	0.071	0.082*	0.071
Round, year, cohort FE			✓		
Male, sibs., mom age, parent educ.	—	✓	✓	✓	✓
N (individuals / person-years, rounded)			1,240 / 4,250		

Correcting for Measurement Error: A Mixture-Simulation Procedure

Back Following Attanasio et al. (2020), I recover synthetic, error-free draws of θ_t, P_t, and I_t in three stages:

1. Estimate each age bin's measurement-error variance from the factor model.
2. Fit a two-component normal mixture to the observed scores and net out that noise, recovering the error-free joint distribution.
3. Draw synthetic error-free values per child and re-estimate the production functions on them.

The mixture allows for skew and thick tails a single normal would miss; correction requires measurement error mean-independent of the true factors.

Is the Bond a Distinct Input, or Just Investment?

Back

	SE Skill (θ_{t+1})			
	[1] Baseline	**[2]** No investment	**[3]** No bond	**[4]** Bond in investment
Investment	0.021	—	0.034*	0.469***
	(0.019)		(0.019)	(0.038)
SE Skill (lag)	0.605***	0.604***	0.659***	0.542***
	(0.049)	(0.049)	(0.038)	(0.046)
Parental Bond (lag)	0.100*	0.105**	—	—
	(0.052)	(0.051)		
Controls		✓		
Observations		3,480		

- Mediation: dropping I_{t+1} leaves the bond coefficient unchanged [2].
- Folding parenting-quality items into I_{t+1}, as in Cunha et al. (2010) and Toppeta (2022), restores a large effect on skill [4].
- Suggests their investment coefficient reflects the bond, not investment narrowly defined.

Parental Time Use (GSS, weekly hours)

Back

	All	Mothers	Fathers
Market work	34.57	24.09	46.67
Active parenting time	11.73	15.16	7.77
Leisure (residual)	121.70	128.75	113.56

- Fathers report nearly twice the market hours of mothers; mothers nearly twice the active time of fathers.
- The market-hours gap is not absorbed one-for-one by active time. Mothers also hold ∼15 more residual hours (which bundles domestic work).
- Both the level and the division of active time track each parent's employment status.

GSS 2005, Cycle 19; parents of children under 14. Residual = 168− market work − active time.

Child-Related Expenditure (SHS, annual $CAD)

Back

	Mean	Std. Dev.
Total child-related expenditure	2,543	3,169
Education (tuition, supplies, books, extracurriculars)	1,118	3,159
Toys	275	416
Clothing and footwear	159	313
Household income	85,595	47,585

- Education is the largest category, over four times toy spending and over seven times clothing.
- Wide dispersion in income (SD ≈$47.6k) carries through to spending.

SHS 1997–2009; households with a child under 14. Rounded.

Investment Measure Items

Back

Parental School Participation (6 items, parent-reported binary): "During this school year, have you. . . "	
1	Spoken to, visited, or corresponded with your child's teacher
2	Visited your child's class
3	Attended a school event (play, sports competition, science fair)
4	Gone on a class trip with your child
5	Helped elsewhere in the school (library, computer room)
6	Attended a parent-school association or advisory committee meeting
Active Parenting Time (3 items, parent-reported ordinal)	
1	How often do you and your child talk about schoolwork or behaviour?
2	How often do you talk about his/her school friends or activities?
3	How often do you check or help with homework?
Outside-School Activities (4 items, child-reported ordinal): "In the past 12 months, how often have you. . . "	
1	Taken part in dance, gymnastics, karate, or similar lessons
2	Taken part in art, drama, or music groups or lessons
3	Played sports WITH a coach or instructor
4	Played sports or physical activities WITHOUT a coach or instructor

Skill Measure Items I

	Hyperactivity/Inattention (7 items)
1	Can't sit still, restless
2	Easily distracted, trouble sticking to an activity
3	Can't concentrate or pay attention
4	Impulsive, acts without thinking
5	Difficulty waiting for turn in games or group activities
6	Cannot settle to anything for more than a few moments
7	Inattentive, difficulty paying attention to someone

Back

	Anxiety/Emotional Disorder (6 items)
1	Unhappy or sad
2	Not as happy as other kids
3	Too fearful or nervous
4	Cries a lot
5	Nervous, highstrung, or tense
6	Trouble enjoying myself

All items child-reported, ages 10–15; NLSCY behavioural scales.

Skill Measure Items II

Back

Pro-social (10 items)		
1	Shows sympathy to someone who made a mistake	All items
2	Tries to help someone who has been hurt	
3	Offers to help clear up a mess someone else made	
4	Tries to stop an argument	
5	Offers to help others having difficulty with a task	
6	Comforts another child who is crying or upset	
7	Helps pick up things another child has dropped	
8	Invites bystanders to join in a game	
9	Helps others who are feeling sick	
10	Encourages others who cannot do things as well	
Conduct Disorder/Aggression (6 items)		
1	Gets into many fights	
2	Assumes accidental hurt was intentional; reacts with anger, fighting	
3	Physically attacks people	
4	Threatens people	
5	Bullies or is mean to others	
6	Kicks or hits other people	

child-reported, ages 10–15; NLSCY behavioural scales.

Bond Measure Items

◂ Back (Bond) ◂ Back (Data)

	Parental Nurturance (7 items): all items begin "My parents. . . "
1	smile at me
2	praise me (say nice things about me)
3	make sure I know I am appreciated
4	listen to my ideas and opinions
5	and I solve a problem together when we disagree
6	speak of the good things I do
7	seem proud of the things I do

All items

	Parental Rejection (7 items): all items begin "My parents. . . "
1	soon forget a rule they have made
2	nag me about little things
3	only keep rules when it suits them
4	threaten punishment more often than they use it
5	enforce rules depending on their mood
6	hit me or threaten to do so
7	get angry and yell at me

child-reported, ages 10–15; NLSCY parenting scales (Lempers et al. 1989). Higher nurturance → less depression and loneliness; rejection → more depression, delinquency, drug use.

Bond Measure Items: Parental Knowledge

◂ Back (Bond) ◂ Back (Data)

Parental Knowledge (4 items): all items begin "My parents. . . "

1	want to know exactly where I am and what I am doing	All items
2	tell me what time to be home when I go out	
3	find out about misbehavior	
4	let me go out any evening I want	

child-reported, ages 10–15; NLSCY parenting scales (Lempers et al. 1989, building on Strayhorn & Weidman; label follows Stattin & Kerr, 2000).

Higher parental knowledge → less delinquency, antisocial behaviour, drug use.

Measurement System Loadings: Skill & Bond

◂ Back (Bond) ◂ Back (Data)

Measure	Ages 10–11		Ages 12–13		Ages 14–15		Invariant	
	λ_t	μ_t	λ_t	μ_t	λ_t	μ_t	λ	μ
Child Socio-Emotional Skills								
Pro-social	1.00[a]	0.00[a]	1.00[a]	0.00[a]	1.00[a]	0.00[a]	1.00[a]	0.00[a]
Conduct Disorder/Physical Aggression	1.15	-0.59	0.90	0.16	1.94	-2.48	1.07	-0.31
Hyperactivity/Inattention	2.50	-4.28	1.83	-2.30	3.52	-6.46	1.93	-2.61
Anxiety and Emotional Disorder	2.21	-3.51	1.45	-1.29	2.73	-4.45	1.58	-1.68
Parental-Child Bond								
Parental Nurturance	1.00[a]	0.00[a]	1.00[a]	0.00[a]	1.00[a]	0.00[a]	1.00[a]	0.00[a]
Parental Knowledge	0.53	1.07	0.38	1.57	0.35	1.64	0.42	1.44
Parental Rejection	0.47	1.53	0.43	1.60	0.49	1.36	0.57	1.18

[a] Marker indicator: loading and intercept fixed for identification, not estimated. Invariant columns constrain λ, μ to be equal across age bins; used for anchoring the latent mean's age profile.

Measurement System Loadings: Investment

Back (Bond) Back (Data)

Measure	Ages 10–11		Ages 12–13		Ages 14–15	
	λ_t	μ_t	λ_t	μ_t	λ_t	μ_t
Outside-School Activities	1.00[a]	0.00[a]	1.00[a]	0.00[a]	1.00[a]	0.00[a]
Parental School Participation	1.38	0.07	1.87	-0.49	1.69	-0.13
Talks About School	2.09	-0.40	2.96	-1.25	2.84	-0.67
Talks About Friends/Activities	2.12	-0.42	2.23	-0.46	2.08	0.03
Checks Homework	2.33	-0.79	3.70	-2.37	3.53	-2.04
Homework Time	0.89	0.05	1.45	-0.37	0.78	0.53

[a] Marker indicator: loading and intercept fixed for identification, not estimated. Investment measures are estimated freely by age group; no age-invariant restriction is imposed.

Latent Factors: Measurement and Moments

Back

Measurement model. For item m on factor $s \in \{\theta, P\}$ at time t:

$$Z_{t,s,m} = \mu_{t,s,m} + \lambda_{t,s,m} \ln s_t + \varepsilon_{t,s,m}$$

With standard restrictions, identifies the means and variances of $\ln \theta_t$, $\ln P_t$.

Bartlett scores.

$$\hat{s}_t^B = \ln s_t + W\varepsilon_t, \qquad W = (\Lambda'\Psi^{-1}\Lambda)^{-1}\Lambda'\Psi^{-1}$$

GLS/BLUE under standard assumptions (Heckman et al., 2013; Croon, 2002).

Which moments are valid:

- Cross-covariances with exogenous X_t preserved: $\mathrm{Cov}(X_t, \hat{s}_t^B) = \mathrm{Cov}(X_t, \ln s_t)$.
- Serial covariances preserved assuming item errors are independent over time.
- Scale inflation: $\mathrm{Var}(\hat{s}_t^B) = \mathrm{Var}(\ln s_t) + W\Psi W'$. Do not use scores for variances; use model-implied variances for standardisation.

Latent Factors: Identification (scalar form)

Back

Single factor S with indicators $Z_m = \mu_m + \lambda_m S + \varepsilon_m$, $\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.

Reference item: $\mu_1 = 0$, $\lambda_1 = 1$.

Off-diagonal covariances ($i \neq j$): $\mathrm{Cov}(Z_i, Z_j) = \lambda_i \lambda_j \mathrm{Var}(S)$.

Recover the latent variance: $\mathrm{Var}(S) = \mathrm{Cov}(Z_1, Z_j)/\lambda_j$ for any $j \neq 1$ (overidentified).

Recover loadings: $\lambda_j/\lambda_k = \mathrm{Cov}(Z_1, Z_j)/\mathrm{Cov}(Z_1, Z_k)$.

Recover unique variances: $\psi_m = \mathrm{Var}(Z_m) - \lambda_m^2 \mathrm{Var}(S)$.

Preferences: Functional Form

Back

Per-period utility, unitary, additively separable, logarithmic:

$$u_t = \alpha_c \ln c_t + \alpha_P \ln P_t + \alpha_{\ell_1} \ln \ell_{1t} + \alpha_{\ell_2} \ln \ell_{2t}, \qquad \sum \alpha = 1$$

Wage Process: Functional Form

Back

For each parent $j \in \{1, 2\}$, the log wage offer in age bin t:

$$\ln w_{jt} = b_{0j} + b_{1j}\,\mathrm{edu}_j + b_{2j}\,\mathrm{age}_{jt} + \varepsilon_{jt}$$

Shocks are i.i.d. across bins and jointly normal:

$$\begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2_{\mathrm{mom}} & \rho_{\mathrm{wage}}\sigma_{\mathrm{mom}}\sigma_{\mathrm{dad}} \\ \rho_{\mathrm{wage}}\sigma_{\mathrm{mom}}\sigma_{\mathrm{dad}} & \sigma^2_{\mathrm{dad}} \end{pmatrix} \right)$$

Parameter Estimates: Skill & Bond Technologies

Back

	Description	Estimate	SE
Skill technology			
a	Intercept	-0.161	0.0038
δ_θ	Age trend	0.076	0.0018
γ_1	Persistence	0.748	0.0019
γ_2	Investment	0.041	0.0005
γ_3	Bond $\rightarrow$ skill	0.179	0.0013
γ_4	Skill $\times$ bond	0.024	0.0003
$\mathrm{sd}(\varepsilon_\theta)$	Shock SD	0.086	0.0026
Bond technology			
$\tilde{a}$	Intercept	0.209	0.0041
δ_P	Age trend	0.128	0.0026
$\tilde{\gamma}_1$	Persistence	0.701	0.0015
$\tilde{\gamma}_2$	Investment	0.088	0.0009
$\tilde{\gamma}_3$	Skill $\rightarrow$ bond	0.068	0.0015
$\tilde{\gamma}_4$	Skill $\times$ bond	0.057	0.0009
$\mathrm{sd}(\varepsilon_P)$	Shock SD	0.234	0.0138

- **Both stocks are persistent**.
- **Investment raises both stocks**, and the two stocks are cross-productive.
- **In SD units:** 1 SD higher initial bond $\Rightarrow$ $+0.425$ SD skill; 1 SD higher initial skill $\Rightarrow$ $+0.118$ SD bond.

Latent Units Model Fit Wage Process

Parameter Estimates: Wage Process

Back

	Description	Estimate	SE
b_{01}	Mom log-wage intercept	0.587	0.0040
b_{11}	Mom education slope	0.081	0.0003
b_{21}	Mom age slope	0.028	0.0001
b_{02}	Dad log-wage intercept	0.564	0.0034
b_{12}	Dad education slope	0.061	0.0002
b_{22}	Dad age slope	0.039	0.0001
σ_{mom}	Mom wage shock SD	0.442	0.0041
σ_{dad}	Dad wage shock SD	0.483	0.0273
ρ_{wage}	Cross-parent shock correlation	0.747	0.0088

Return to schooling: $\sim$6.3% per year for fathers, $\sim$8.5% for mothers. Fathers face $\sim 1.1\times$ the wage risk of mothers. Shocks strongly correlated across parents (assortative mating, shared local labour market).

Model Fit: Child Skill and Parental Bond

◂ Back (Prefs) ◂ Back (Tech)

	Data		Model	
Child Age	Mean	SD	Mean	SD
Panel A: Child Skill (θ_t)				
10–11	2.600	0.129	2.600	0.105
12–13	2.561	0.106	2.557	0.150
14–15	2.505	0.112	2.567	0.190
Panel B: Parental Bond (P_t)				
10–11	3.158	0.241	3.159	0.194
12–13	3.101	0.254	3.048	0.299
14–15	3.028	0.248	3.064	0.359

Model Fit: Labour Supply and Wages

◂ Back (Prefs) ◂ Back (Tech)

	Mother				Father			
	Data		Model		Data		Model	
Child Age	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Panel A: Hours Worked (conditional on employment)								
10–11	30.40	12.79	32.67	14.97	42.21	11.29	41.41	13.46
12–13	31.01	13.08	32.59	15.07	42.77	10.68	41.87	13.39
14–15	32.11	12.43	32.90	15.16	42.84	11.05	42.33	13.26
Panel B: Employment Rate								
10–11	0.829	—	0.847	—	0.967	—	0.960	—
12–13	0.863	—	0.844	—	0.966	—	0.964	—
14–15	0.895	—	0.845	—	0.961	—	0.966	—
Panel C: Accepted Wage (log, conditional on employment)								
10–11	2.789	0.641	2.845	0.479	3.072	0.485	3.004	0.532
12–13	2.829	0.646	2.918	0.482	3.098	0.460	3.095	0.529
14–15	2.833	0.618	2.976	0.473	3.115	0.495	3.166	0.540

Terminal Value

Back

- **Skill terminal** λ_θ: freely estimated inside the SMM. Captures the weight parents place on the child's socio-emotional skill at the end of childhood.
- **Bond terminal**: none. No post-childhood problem pins P, so the bond pays off only through flow utility while the child is at home.
- Terminal value: $V(\theta_{T+1}) = \lambda_\theta \ln \theta_{T+1}$.

Derivation

Terminal Value: Derivation

Back

- **Skill**: suggestive benchmark, not an imposed restriction. Child enters adulthood in a stationary environment with log utility:
  - Adult outcomes (earnings, broadly) $\ln y^* \approx \psi \ln \theta_{T+1}$.
  - Consume all income $\Rightarrow V^{\text{child}} = \ln y^* + \beta V^{\text{child}}$.
  - Fixed point: $V^{\text{child}} \approx \dfrac{\psi}{1-\beta} \ln \theta_{T+1}$.
  - Parents weight it by κ; κ, ψ not separately identified $\Rightarrow$ estimate $\lambda_\theta \equiv \kappa\psi/(1-\beta)$.

- **Bond**: no post-childhood problem pins $P \Rightarrow$ no terminal value; the bond enters only flow utility.

$$\boxed{V(\theta_{T+1}) = \lambda_\theta \ln \theta_{T+1}}$$

Why Do Parents Invest in the Bond?

Back Two reasons:

1. *Flow utility* $u_t(\ldots, P_t)$: parents enjoy the relationship each period.
2. *Skill spillover* $\gamma_3 \ln P_t + \gamma_4 \ln \theta_t \ln P_t$: the bond raises θ_{t+1}.

Even a parent who places little direct weight on the bond still has a **skill-driven** reason to invest in it.

Full Laws of Motion

Back

$$\ln \theta_{t+1} = \ln A(t) + \gamma_1 \ln \theta_t + \gamma_2 \ln I_t + \gamma_3 \ln P_t + \gamma_4 \ln \theta_t \ln P_t + \varepsilon_{t,\theta}$$

$$\ln P_{t+1} = \ln \tilde{A}(t) + \tilde{\gamma}_1 \ln P_t + \tilde{\gamma}_2 \ln I_t + \tilde{\gamma}_3 \ln \theta_t + \tilde{\gamma}_4 \ln \theta_t \ln P_t + \varepsilon_{t,P}$$

$$\ln I_t = \alpha_e \ln e_t + \alpha_{m_1} \ln m_{1t} + \alpha_{m_2} \ln m_{2t} + \alpha_{12} \ln m_{1t} \ln m_{2t}$$

with $A(t) = e^{a+\delta_\theta t}$, $\tilde{A}(t) = e^{\tilde{a}+\delta_P t}$.

- θ_t drives the terminal value; P_t enters flow utility directly.
- The reduced-form investment interactions are n.s.; the skill $\times$ bond interaction is significant and is included (γ_4, $\tilde{\gamma}_4$).

Alternative Specification: α_θ in Flow Utility

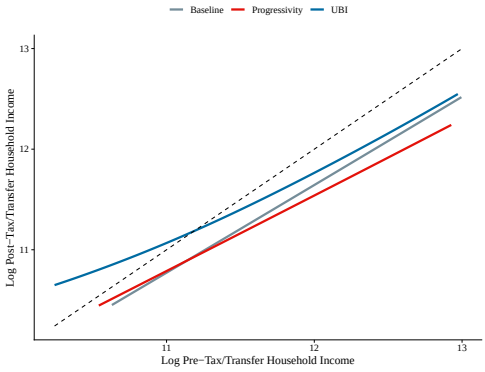
Back

Admit child skill into flow utility with weight α_θ, and construct the skill terminal under the same perpetuity convention used for the bond, so α_θ and α_P are estimated jointly.

- There, the household values a more capable child ($\alpha_\theta = 0.117$) over three times as much as a closer relationship ($\alpha_P = 0.036$, against 0.068 in the baseline).
- The split between α_θ and α_P is recovered **far less precisely**, relative to its own magnitude, than α_P and λ_θ are in the baseline, because the two envelope conditions take the same recursive form.
- This is the identification cost that the baseline exclusion of θ_t from flow utility avoids.

Tax Systems: Mechanical Mapping

Back



Pre-tax to post-tax household income under the baseline system, the \$24k UBI (higher $\mathcal{X}_2$), and the doubled-progressivity system ($\mathcal{X}_3 = 0.245$, $\mathcal{X}_1$ readjusted), before any behavioural response.

SMM Algorithm

Back Choose Ω to minimise the distance between simulated and empirical moments:

$$Q(\Omega) = \big[m^{\text{data}} - \widehat{m}(\Omega)\big]' W \big[m^{\text{data}} - \widehat{m}(\Omega)\big], \qquad W = \operatorname{diag}(\widehat{\Sigma}_B)^{-1}$$

1. Guess Ω^k; draw wage offers for mother and father each period.
2. Solve the household problem $\rightarrow$ optimal policy functions.
3. Simulate the joint distribution of (θ_{t+1}, P_{t+1}) and all choices.
4. Compute $\widehat{m}(\Omega^k)$, evaluate Q, update via Nelder–Mead.
5. Repeat until $\Omega^{k+1} \approx \Omega^k$.

$\widehat{\Sigma}_B$ is the block-bootstrap variance of the empirical moments ($B = 1{,}000$). SEs from 500 bootstrap replications of the estimator.

SMM: Moment Detail and Weighting

Back

- $W = \mathrm{diag}(\widehat{\Sigma}_B)^{-1}$: draw 1,000 block-bootstrap samples, compute the moment vector in each, set W to the inverse of the diagonal of their sample covariance.
- Objective minimised with the Nelder–Mead simplex at each iteration.
- Standard errors from 500 bootstrap replications of the full estimator.
- Order condition: $\sim$150 moments for 30 parameters (after two adding-up restrictions).

Identification: Parameter Blocks → Moments

Back

Parameter block	Primary identifying moments
Skill & bond technologies γ_1–γ_4, $\tilde{\gamma}_1$–$\tilde{\gamma}_4$	Measurement-error-corrected production function estimates; age-bin variances, autocovariances, and cross-lag covariances of $\ln \theta_t$, $\ln P_t$; covariance of latent states with investment
Productivity $a, \tilde{a}, \delta_\theta, \delta_P$	Age profile of latent means (age-invariant anchoring)
Aggregator $\alpha_e, \alpha_{m_1}, \alpha_{m_2}, \alpha_{12}$	SHS expenditure-to-income ratio by employment; GSS level & division of parental time by couple type
Leisure $\alpha_{\ell_1}, \alpha_{\ell_2}$	NLSCY employment states & hours by age bin
Consumption α_c	After-tax income moments; active-time–work trade-off; SHS income elasticity of the expenditure share
Bond weight α_P & skill terminal λ_θ	Age-bin profiles of employment, hours, and household income, jointly
Wage offers	Accepted-wage mean, education covariation, age profile; dispersion; spousal correlation

Latent Units: What Can Be Compared

◂ Back (Prefs) ◂ Back (Tech)

Skill and the bond have no natural unit: the marker item (pro-social, nurturance) sets the scale.

Example. Normalise the bond on rejection (loading 0.57) instead of nurturance:

$$\alpha_P \ln P_t = \frac{\alpha_P}{0.57} \ln P_t^{\text{rej}} \quad \Rightarrow \quad \alpha_P^{\text{rej}} = 0.196 \text{ after rescaling to sum to one}$$

On parental knowledge (loading 0.42): 0.249. Baseline: 0.122.

Unit-free

- Persistence γ_1, $\tilde{\gamma}_1$
- Aggregator weights α_e, α_{m_1}, α_{m_2}
- Consumption vs. leisure weights

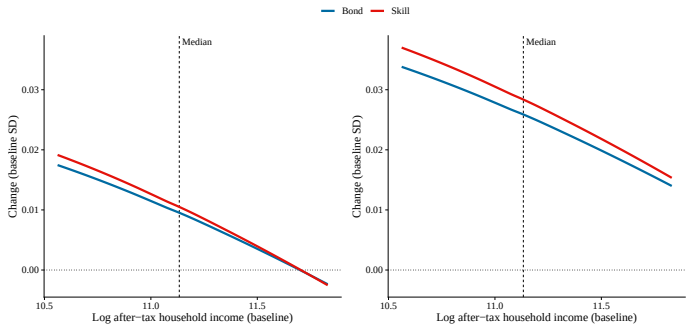
Depend on units

- γ_2 vs. $\tilde{\gamma}_2$; γ_3 vs. $\tilde{\gamma}_3$
- α_P vs. α_c, α_{ℓ_j}

$\Rightarrow$ Compare through counterfactuals in SD units (Del Bono et al., 2022).

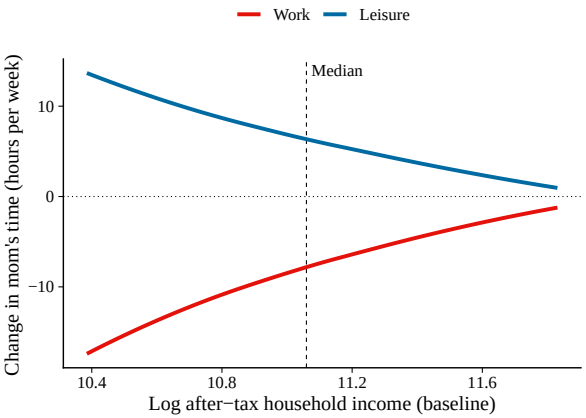
Progressivity: Average vs. Marginal Rate Held Fixed

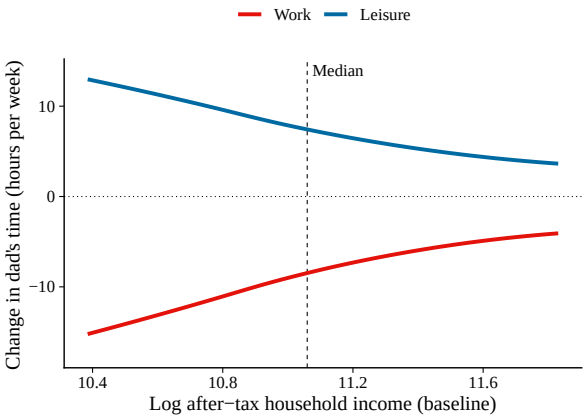
Back



- Left: median <u>average</u> rate fixed (baseline). Right: median <u>marginal</u> rate fixed (Agostinelli et al., 2024), so $\mathcal{X}_1$ $\sim$16% higher.
- Right: larger gains at every income level. Gains still fall with income; skill moves more than the bond.

Back

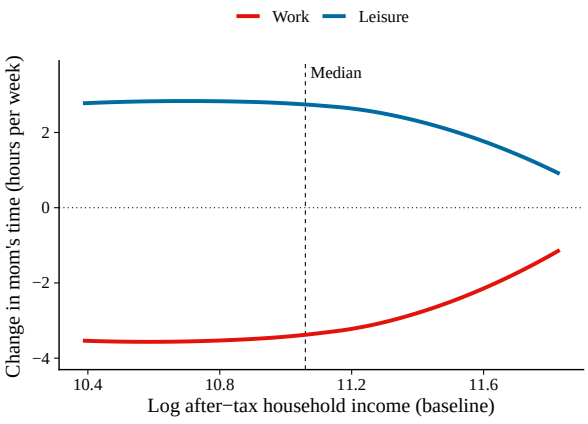


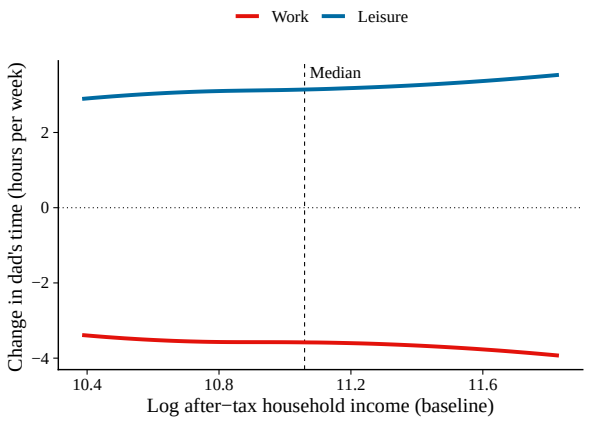


- Dad's time concentrated at low income (fades out), Mom's falls steadily with income

Progressivity

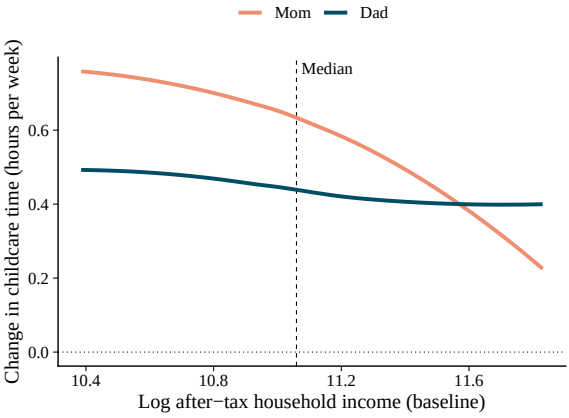
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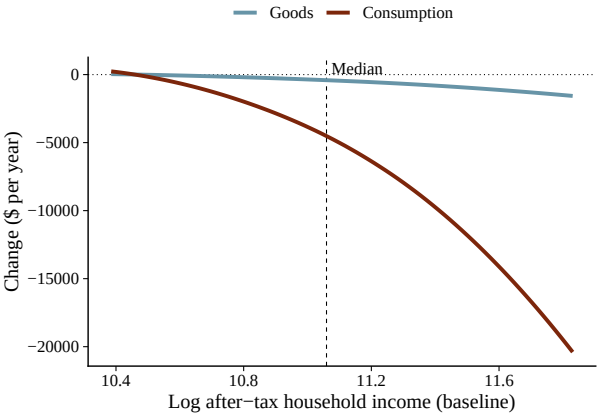




- Same direction as UBI, but asymmetric above the median: moms ramp work back up, dads keep cutting.

Back





- Goods & consumption fall as income rises, consumption far more than goods.

Counterfactual Results: Summary

Back

	Relationship Motive			Terminal Skill Wedge		Both	Initial Endowments		Tax-and-Transfer	
	$\alpha_P{=}0$	$\alpha_P{\downarrow}50\%$	$\alpha_P{\uparrow}50\%$	$\chi_\theta{\downarrow}50\%$	$\chi_\theta{\uparrow}50\%$	$\alpha_P, \chi_\theta{\uparrow}50\%$	$\theta_0{\uparrow}1$ SD	$P_0{\uparrow}1$ SD	UBI \$24k	$\mathcal{X}_3$ dbl
Panel A: Terminal outcomes (skill and bond in SD, adult income in %)										
Child Skill θ_{T+1}	−0.100	−0.050	0.050	−0.262	0.155	0.197	0.409	0.425	0.055	0.010
Parental Bond P_{T+1}	−0.090	−0.045	0.045	−0.242	0.143	0.181	0.118	0.418	0.050	0.009
Adult Income	−1.73	−0.87	0.89	−4.49	2.75	3.51	7.42	7.73	0.96	0.18
Panel B: Parental labour market (%)										
Mother Hours	2.53	1.33	−1.48	6.68	−5.90	−7.60	−0.05	−0.08	−28.68	−11.98
Father Hours	−0.13	−0.07	0.07	−0.41	0.24	0.29	0.00	0.00	−21.34	−8.91
Household Income	0.67	0.35	−0.40	1.75	−1.63	−2.11	−0.01	−0.02	9.53	−8.18
Panel C: Child expenditure and active time (%)										
Child Expenditure	−16.31	−8.58	9.59	−43.62	39.00	50.26	0.33	0.49	9.68	−8.06
Mother Active Time	−17.03	−8.96	10.01	−44.83	40.16	51.80	0.35	0.53	10.20	4.21
Father Active Time	−17.74	−9.36	10.52	−46.05	42.29	54.79	0.37	0.56	13.13	5.38

Skill and bond: change in mean log stock at 14–15, in SD of its baseline distribution. Adult income anchored with income-table column [2] (0.943 per unit of $\ln\theta$).

Tax-and-Transfer Results by Income Quartile

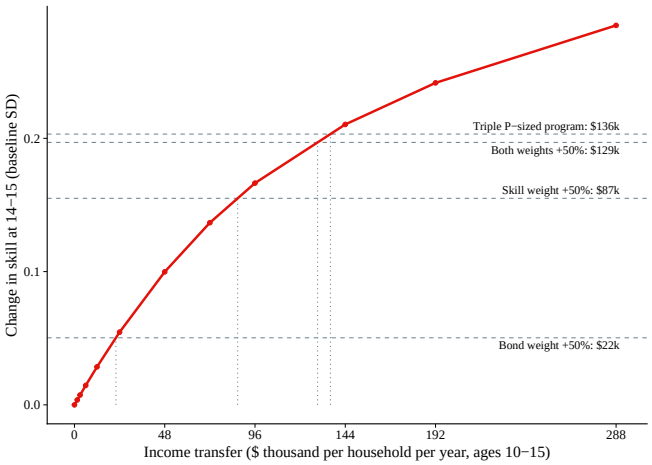
Back

	UBI $24k				$\mathcal{X}_3$ Doubled			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Panel A: Terminal outcomes (skill and bond in SD, adult income in %)								
Child Skill θ_{T+1}	0.069	0.057	0.051	0.041	0.014	0.012	0.010	0.006
Parental Bond P_{T+1}	0.063	0.052	0.046	0.038	0.013	0.010	0.009	0.006
Adult Income	1.21	1.01	0.89	0.73	0.25	0.20	0.17	0.11
Panel B: Parental labour market (%)								
Mother Hours	−32.55	−28.91	−26.59	−25.27	−10.93	−11.72	−12.14	−13.71
Mother Employment Rate	−11.64	−11.54	−11.36	−12.74	−3.05	−4.07	−4.64	−6.60
Father Hours	−29.01	−22.96	−19.71	−15.10	−9.74	−9.11	−8.82	−8.12
Father Employment Rate	−8.79	−5.32	−3.96	−1.98	−1.86	−1.42	−1.31	−0.78
Household Income	12.17	10.20	9.09	7.71	−5.14	−7.16	−8.51	−10.61
Panel C: Child expenditure and active time (%)								
Child Expenditure	12.36	10.36	9.22	7.83	−5.02	−7.04	−8.39	−10.50
Mother Active Time	14.22	11.03	9.35	6.69	4.71	4.40	4.20	3.58
Father Active Time	15.63	13.61	12.24	10.80	5.15	5.31	5.39	5.71

Quartiles of baseline after-tax household income (Q1 lowest), membership fixed across scenarios. Skill and bond in SD of the pooled baseline distribution.

How Large a Transfer Matches Each Experiment?

Back



Annual transfer, ages 10–15, with the same skill gain at 14–15

- Bond weight +50%: $22k
- Skill weight +50%: $87k
- Both weights: $129k
- Triple P-sized program (bond +0.2 SD): $136k

Gains from a transfer flatten with its size: parents spend most of it on consumption and leisure.

A Triple P-Sized Program and Its Cash Equivalent

Back

Each experiment leaves the bond **0.2 SD** higher at ages 14–15.

	Bond	Skill	Adult	Time (hr/wk)		Spending	Transfer, same skill gain ($000/yr)		
	(SD)	(SD)	Income	Mother	Father	($/yr)	All	Q1	Q4
Triple P-sized bond shock									
Bond shock, 0.48 SD at ages 10–11	0.200	0.203	3.62%	0.04	0.02	13	136.1	101.9	185.2
Preference changes sized to the same bond gain									
Bond weight $\alpha_P \uparrow$ 213%	0.200	0.221	3.94%	7.82	4.57	2,716	159.7	120.9	219.8
Skill weight $\chi_\theta \uparrow$ 77%	0.200	0.217	3.88%	8.93	5.20	3,137	154.8	117.4	212.9
Both, α_P and $\chi_\theta \uparrow$ 57%	0.200	0.218	3.89%	8.71	5.07	3,054	155.9	118.2	214.4

- Matching the bond shock takes $136k a year; the preference changes take $155–160k.
- Preference changes work through time: mothers +8 to 9 hrs/wk, fathers +5. The bond shock barely moves inputs.

Mechanisms: The Relationship Motive

Back

Counterfactual: set $\alpha_P = 0$, so parents receive no direct utility from the relationship.

Baseline: parents invest in the bond for two reasons.

$$u_t = \alpha_c \ln c_t + \alpha_{\ell_1} \ln \ell_{1t} + \alpha_{\ell_2} \ln \ell_{2t} + \alpha_P \ln P_t, \qquad V = \lambda_\theta \ln \theta_{T+1}$$

1. They enjoy it each period ($\alpha_P \ln P_t$).
2. It raises skill: $\ln \theta_{t+1} = a + \gamma_1 \ln \theta_t + \gamma_2 \ln I_t + \gamma_3 \ln P_t + \gamma_4 \ln \theta_t \ln P_t + \varepsilon_{t,\theta}$.

Under $\alpha_P = 0$ only the **investment** motive (2) survives, through γ_3 and γ_4.

The Relationship Motive: Results

Back

Removing the enjoyment motive cuts investment by about a sixth. (% change from baseline)

Mother active time	−17.0
Father active time	−17.7
Child expenditure	−16.3
Mother hours	+2.5
Father hours	−0.1
Household income	+0.7
Child skill θ_{T+1} (SD)	−0.100
Parental bond P_{T+1} (SD)	−0.090

- Freed time goes mostly to **leisure**; consumption rises as child spending falls.
- Skill and the bond fall by only 0.100 and 0.090 SD: both are persistent, and the bond still raises skill through γ_3, γ_4.
- α_P ±50%: investment −9% / +10%; skill ±0.050 SD, symmetric.

About a sixth of baseline investment reflects enjoyment of the relationship, and the rest its return to the child.

Mechanisms: The Weight on the Child's Future

Back

Counterfactual: scale the skill terminal by χ_θ ($= 1$ at baseline), so parents care more or less about the child's future success.

$$V(\theta_{T+1}) = \chi_\theta \, \lambda_\theta \ln \theta_{T+1}$$